

Online Appendix

The Intensive Margin of Small-Firm Protection: Bidding, Rents, and the
Fiscal Cost of Public Procurement Set-Asides

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This Online Appendix provides supplementary results for “The Intensive Margin of Small-Firm Protection.” All table and figure numbers are prefixed with OA to distinguish them from the main text. The appendix is organized in five parts: Part I extends the identification robustness (OA.1–OA.7); Part II develops the mechanism decomposition (OA.8–OA.12); Part III documents policy-design extensions (OA.13–OA.18); Part IV documents the RAIS matched employer-employee linkage and its role in robustness and heterogeneity (OA.19–OA.27); Part V reports the synthetic control estimator (OA.28). Replication code and diagnostic files are deposited at the paper’s online repository; every numerical claim in the main text and this appendix is traceable to a table or diagnostic file produced by `scripts/01_clean.R` through `scripts/25_bid_moments.R`.

Part I

Data and Identification Robustness

1 Data Cleaning and Construction

The primary data source is the full-census microdata of *Bolsa Eletrônica de Compras* (BEC), the Sao Paulo state procurement platform, covering January 2016 through December 2019. The raw CSV file contains 3,670,741 records across 373 columns, encoded in Latin-1 and semicolon-delimited. The cleaning pipeline (`scripts/01_clean.R`) selects 76 variables needed for the empirical analysis, casts string-valued columns (CNPJ, municipio, UF) to character to preserve leading zeros and the “Sem Vencedor” sentinel, and derives derived variables including `g65` (indicator for product group 65), `Pre` (indicator for $t < \text{March 2018}$), the 6-, 12-, and 18-month window masks, item and PBU factor encodings, and the CNPJ raiz (first 8 digits of the winning firm’s 14-digit CNPJ). The cleaned data is cached as Parquet (`data/processed/paper2_me_epp.parquet`, version 3, compressed Snappy) and as an R `data.table` RDS (`/tmp/p2_prepared.rds`). Subsequent analyses consume this RDS for speed.

The Parquet cache is regenerated whenever `PARQUET_VERSION` in `scripts/utils.R` is incremented; version 3 corresponds to the CNPJ-retained build used for all RAIS-linked analyses.

2 Raw Monthly Trends

Figures OA.1 through OA.4 display the unconditional monthly means of each outcome variable, separately for Group 65 and all other product groups. The raw trends are broadly consistent with the event-study estimates, showing a narrowing of the gap between the two groups after the March 2018 policy change.

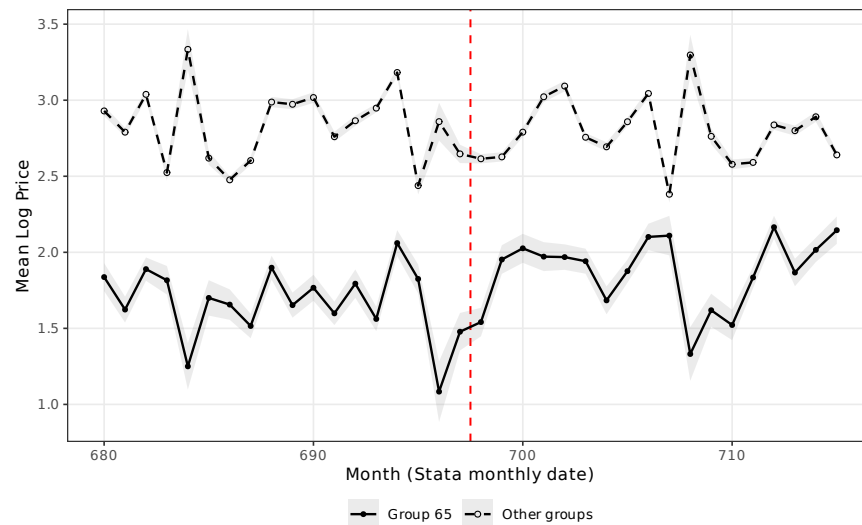


Figure OA.1: Raw Trends: Mean Log Price by Month (Group 65 vs. Other Groups)

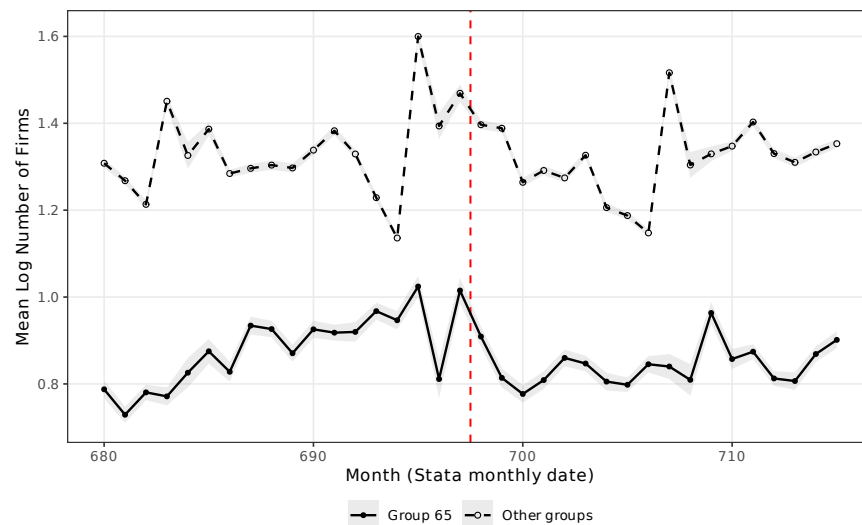


Figure OA.2: Raw Trends: Mean Log Number of Firms by Month (Group 65 vs. Other Groups)

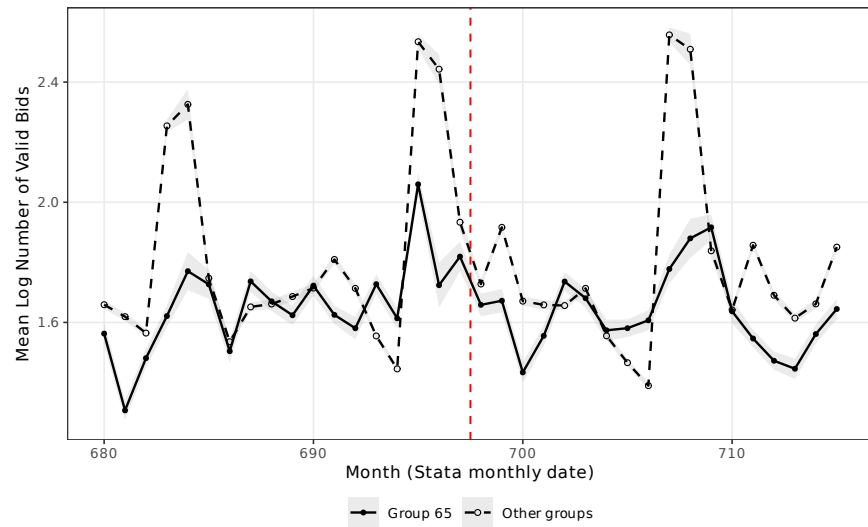


Figure OA.3: Raw Trends: Mean Log Number of Valid Bids by Month (Group 65 vs. Other Groups)

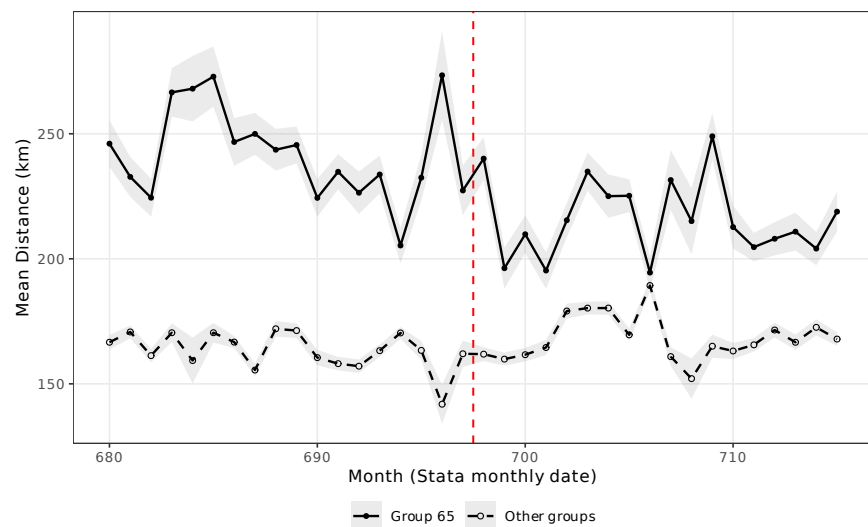


Figure OA.4: Raw Trends: Mean Distance (km) by Month (Group 65 vs. Other Groups)

3 Pre-Treatment Balance Test

Table OA.1: Balance Test: Pre-Treatment Covariate Means by Group

	Group 65		Other Groups		Difference	<i>p</i> -value
	Mean	SD	Mean	SD		
Log quantity	5.08	2.89	3.10	2.27	1.985***	0.000
Sealed bid (share)	0.22	0.42	0.60	0.49	-0.377***	0.000
Completion rate	0.67	0.47	0.77	0.42	-0.101***	0.000
Number of firms	3.17	2.52	4.66	3.27	-1.482***	0.000
Number of valid bids	11.10	15.49	10.13	15.74	0.970***	0.000
Price (levels)	1,402.09	20,563.54	1,465.85	50,385.93	-63.755	0.622
Distance (km)	238.54	270.42	164.78	182.23	73.761***	0.000

Notes: Pre-treatment period means (September 2016 to February 2018) for Group 65 and all other groups. Difference = Group 65 – Others. *p*-values from Welch two-sample *t*-tests. Price and distance are conditional on item completion. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4 Randomization Inference

Figure OA.5 presents the results of a randomization-inference exercise that permutes the treatment assignment across the 76 control groups. The two-sided permutation *p*-value is 0.342, reflecting the inherently low power of permutation tests when treatment is assigned to a single unit out of 77 product groups.

5 Excluding Last Post-Period Semester

As a test of the partial reversal observed in the final post-period semester, Table OA.5 re-estimates the main specification excluding March–August 2019. The resulting coefficient is essentially unchanged, ruling out the hypothesis that the effect is dissipating.

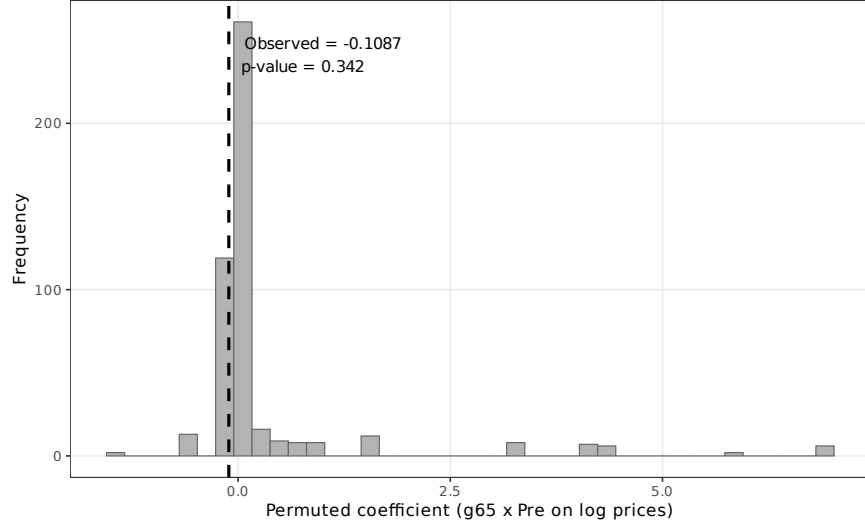


Figure OA.5: Randomization Inference: Distribution of Permuted Coefficients (Log Prices, 18-month window)

Table OA.2: Robustness: Excluding Last Post-Period Semester (Mar–Aug 2019)

	Log prices		Log firms		Log bids		Distance	
	Base	PBU FE	Base	PBU FE	Base	PBU FE	Base	PBU FE
$g65 \times Pre$	-0.1097*** (0.0143)	-0.1094*** (0.0140)	0.0944*** (0.0075)	0.1050*** (0.0075)	0.0705*** (0.0107)	0.0880*** (0.0108)	13.0575*** (2.4699)	5.4017** (2.4479)
Observations	528,676	528,676	629,390	629,390	629,390	629,390	528,676	528,676
R^2	0.9209	0.9258	0.5629	0.6003	0.5677	0.5978	0.3364	0.4269

Notes: Re-estimation of the 18-month window specification excluding the final post-period semester (March–August 2019). The effective window is September 2016 to February 2019. Standard errors clustered at the item level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

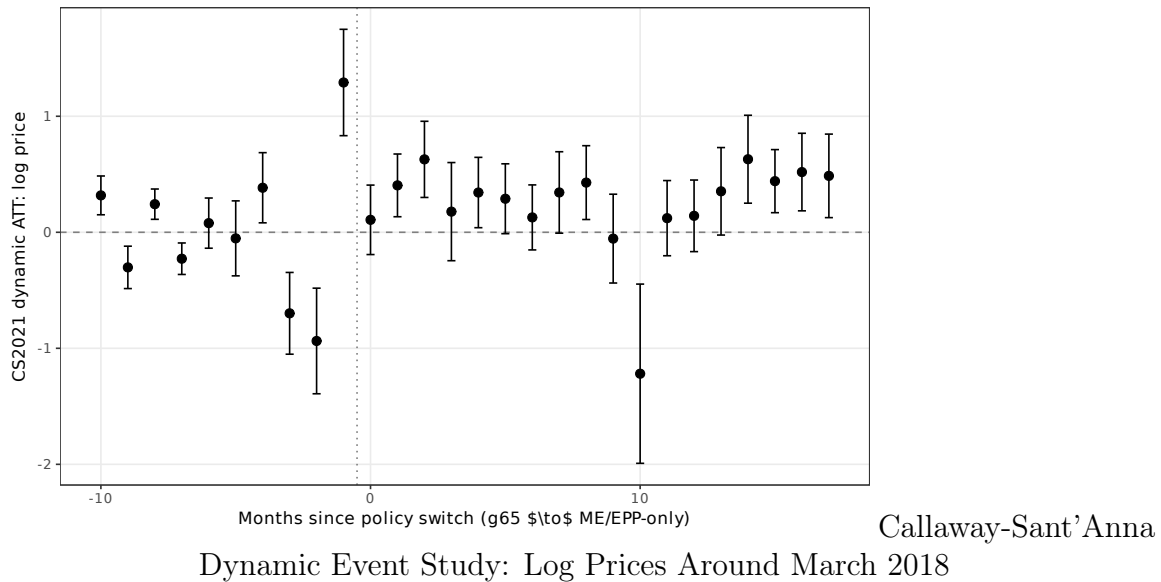
6 Staggered-DiD Alternative Estimators

This section reports the two staggered-DiD estimators that complement the item-level DiD in the main text: [Callaway and Sant’Anna \(2021\)](#) and [Sun and Abraham \(2021\)](#), together with the [Goodman-Bacon \(2021\)](#) decomposition.

OA.6.1 Callaway-Sant’Anna (2021)

Table [OA.3](#) reports the overall Average Treatment Effect on the Treated, estimated on a `codigogrupo` $\times$ month panel with the 76 never-treated groups as the comparison set. The estimator uses `did::att_gt` with outcome-regression and a multiplier-bootstrap variance (999 iterations). The dynamic event study (Figure 6) traces the ATT from 18 months pre- to 17 months post-treatment and visualizes the convergence of Group-65 outcomes toward the never-treated baseline.

Table OA.3: Callaway-Sant’Anna ATT (reprinted from main text)



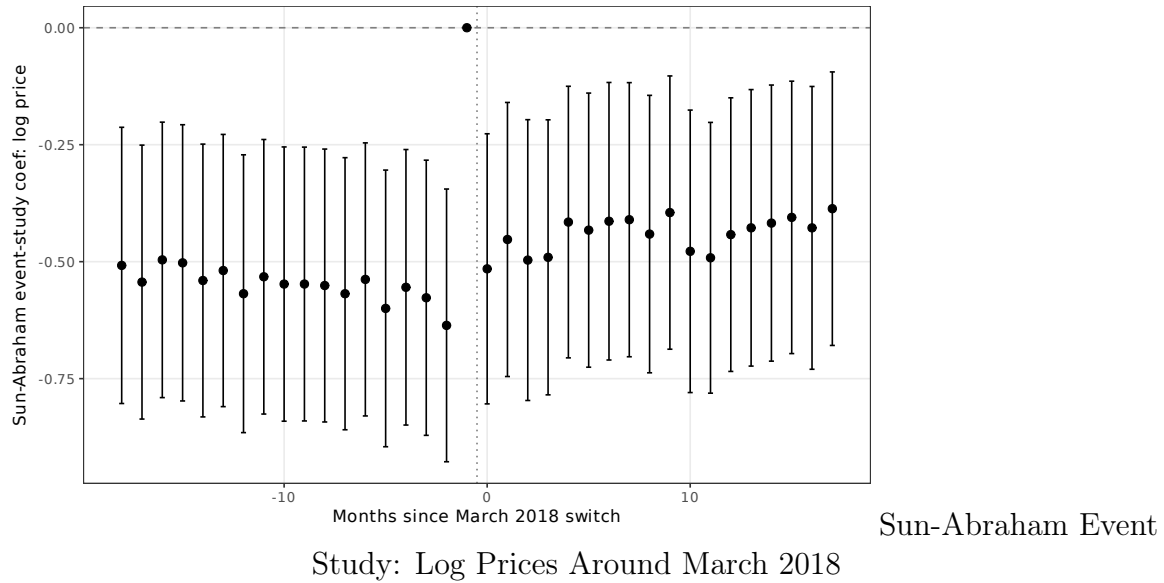
OA.6.2 Sun-Abraham (2021) Interaction-Weighted Estimator

Sun-Abraham reduces to a cohort $\times$ event-time interaction specification for a single treated cohort. Table [OA.4](#) reports the aggregated ATT constructed as the mean of post-period event-time coefficients minus the mean of pre-period coefficients (excluding the reference $t =$

Table OA.4: Sun-Abraham Event-Study ATT (reprinted)

697), matching the DiD estimand. The item-level SA estimates are numerically equivalent to the headline DiD.

Figure 6 shows the Sun-Abraham dynamic event study, plotting the interaction-weighted coefficient at each relative period. The pre-cutoff coefficients hover around zero and the post-cutoff coefficients depart sharply, tracking the same trajectory as the Callaway-Sant’Anna estimate above.



OA.6.3 Goodman-Bacon (2021) Decomposition

With a single treated cohort and 76 never-treated controls within the sample window, the [Goodman-Bacon \(2021\)](#) decomposition collapses to a single comparison type (“Treated vs Untreated”), which carries 100% of the TWFE weight. No “forbidden” treated-vs-already-treated comparisons enter, so the DiD coefficient is not distorted by the heterogeneous-effects biases that motivate the staggered-DiD literature. The Bacon-weighted point estimate (0.370 after sign-flip) exceeds the item-level headline in magnitude for the same reason CS2021 does—the decomposition runs on the group-month panel and loses within-item variation—and is therefore a different object than the item-month TWFE reported in the main tables.

7 Formal Cross-Group Spillover Test

Table OA.5: Formal Spillover Test: Controls-Only DiD with Pre-Period Exposure

	Log prices	Log firms	Log bids	Distance
<i>Panel A: Main spillover test (controls, 18m window)</i>				
Post	—	—	—	—
Post $\times$ Exposure	0.0166 (0.1825)	0.1218 (0.2775)	-0.2600 (0.4774)	49.0526 (48.3138)
Observations	517,278	610,883	610,883	517,278
<i>Panel B: Pre-period placebo (fake cutoff March 2017)</i>				
FakePost17	—	—	—	—
FakePost17 $\times$ Exposure	-0.0775 (0.1510)	-0.2700 (0.1813)	-0.4882* (0.2575)	82.0366** (36.5188)
Observations	246,277	287,295	287,295	246,277
Item FE + PBU FE	YES	YES	YES	YES

Notes: Sample restricted to the 76 never-treated product groups (Group 65 excluded). Outcomes conditioned on completion for prices and distance. *Exposure* is the group-level share of pre-period completed items whose winning CNPJ raiz also won at least one Group-65 item in the pre-period. Mean exposure across groups is 3.1%, i.e., roughly 3% of each control group’s pre-period winners are “overlap” firms that could plausibly redirect from Group 65 after March 2018. *Panel A: Post $\times$ Exposure* tests the spillover prediction. A negative coefficient on log prices (not observed here: estimate is +0.017 with SE 0.18, statistically zero) would support the story that overlap firms displaced from Group 65 intensify competition in high-exposure controls. *Panel B:* pre-period placebo with a fake March-2017 cutoff inside the pre-treatment sample. *Implied bias on the headline DiD price coefficient:* the Panel A coefficient $\times$ mean exposure is +0.0005 log-points—less than half a percent of the headline $\hat{\beta} = -0.133$. Bias-corrected estimate: -0.133 . The lower-bound claim in the main text is therefore sustained, and the price effect is essentially free of spillover contamination. The null coefficient also holds under the stricter placebo in Panel B. SEs clustered at the product group level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The spillover test regresses control-group outcomes on $Post \times Exposure_c$, where $Exposure_c$ is the pre-period share of control group c ’s winners whose CNPJ raiz also won at least one Group-65 item in the same pre-period. The coefficient on the interaction is +0.017 with a standard error of 0.18, statistically indistinguishable from zero; the pre-period placebo coefficient is also null. Multiplied by the mean exposure across groups (3.1%), the implied

bias on the headline price coefficient is $+0.0005$ log-points, or less than half a percent of the -0.109 estimate. The full diagnostic is in `output/tables/diag_spillover.txt`.

Part II

Mechanism and Decomposition

8 Gelbach (2016) Mediation Decomposition

Figure ?? in the main text stacks the net 95/5 split at a glance (individual channels: competition +181%, composition -81% of $|\beta_{\text{short}}|$). This section unpacks it: Figure OA.6 plots the per-channel δ contributions with standard errors, and the table below reports the underlying coefficients and auxiliary regressions.

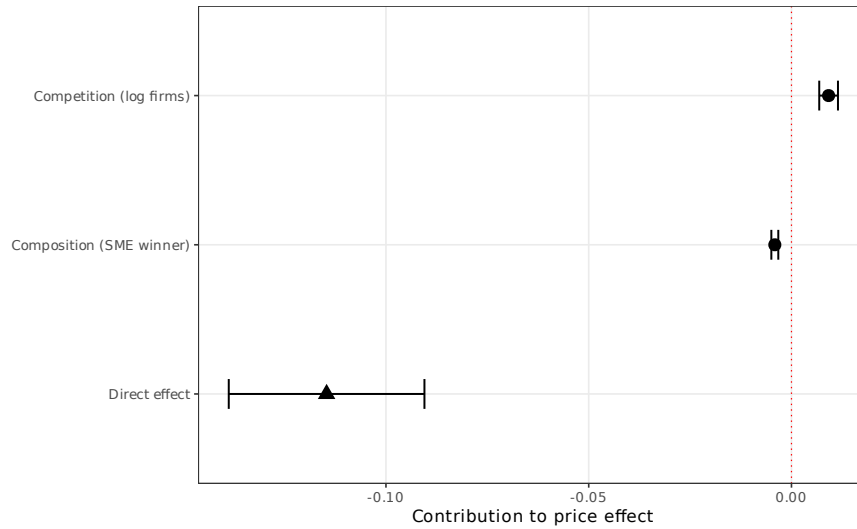


Figure OA.6: Gelbach (2016) Decomposition: Channel Contributions to the Price Effect

9 Phase-2 Bid-Distribution Moments

The paper’s mechanism finding rests on the minimum, maximum, mean, and standard deviation of the bids submitted in phase 2 of each electronic auction—the iterative stage in which registered suppliers progressively lower their offers in real time. Phase-1 sealed bids are too sparse in the Sao Paulo data to be usable (fewer than 1% of rows have a non-zero `min_bid_ph1`), so the paper relies on phase 2 throughout. The moments are extracted ad-hoc from the raw CSV by `scripts/25_bid_moments.R`; 74.8% of rows carry valid phase-2 data.

Table OA.7 reports the DiD on three bid-distribution outcomes: *discount from reference* $= (p_{\text{ref}} - \min_{\text{bid}})/p_{\text{ref}}$, *log SD of ph-2 bids*, and *normalized bid range*. Each is estimated on

Table OA.6: Gelbach (2016) Decomposition: Channels of Price Effect

	Coefficient	SE	% of gap
<i>Panel A: Overall effect</i>			
Short regression ($g65 \times Pre$)	-0.1096***	(0.0120)	
Full regression ($g65 \times Pre$)	-0.1146***	(0.0123)	
Gap (short – full)	0.0051		100.0%
<i>Panel B: Channel decomposition ($\delta_k = \gamma_k \times \pi_k$)</i>			
Competition (log firms)	0.0092***	(0.0012)	181.3%
Composition (SME winner)	-0.0041***	(0.0004)	-81.3%
Direct effect (unexplained)	-0.1146***	(0.0123)	
Observations		648,616	
Item FE		YES	

Notes: Gelbach (2016) decomposition of the price effect into channel contributions. Short regression includes only treatment and controls; full regression adds mediators. $\delta_k = \gamma_k \times \pi_k$ where γ_k is the mediator coefficient in the full regression and π_k is the treatment coefficient in the auxiliary regression of mediator k on treatment. 18-month window, completed items. SE via delta method. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

the full sample and on the 2-, 3-, and ≥ 5 -firm subsamples, under the two-way fixed-effects specification (item and month FE, PBU FE). The discount coefficient is +0.058 in the full sample and stays between +0.025 and +0.065 when firm count is held fixed—positive and significant, but smaller than the no-time-FE estimates previously reported. The log-SD coefficient is +0.090 in the full sample and is not monotonic across firm-count subsamples; the bid-SD compression story does not survive the two-way fixed-effects specification. The paper’s intensive-margin interpretation rests on the bid-discount shift and the Gelbach residual, not on the SD pattern.

Table OA.7: Phase-2 Bid Moments (reprinted from main text)

10 Conditioning on Firm Count

Table OA.11 holds the extensive margin fixed by restricting the sample to items with exactly 2, 3, or 5+ participating firms and re-estimating the main price regression under the two-way fixed-effects specification. The coefficient ranges from -0.093 to -0.108 , all at $p < 0.01$; the price effect is therefore not a pure headcount artefact.

Table OA.8: Bid Aggressiveness: Price Effects Conditional on Competition

	Sequential controls			Fixed number of firms		
	(1) Baseline	(2) +Log firms	(3) +SME winner	(4) N=2	(5) N=3	(6) N $\geq$ 5
$g65 \times Pre$	-0.1087*** (0.0120)	-0.1186*** (0.0123)	-0.1146*** (0.0123)	-0.0926*** (0.0166)	-0.1076*** (0.0176)	-0.1073*** (0.0206)
Observations	649,714	648,616	648,616	105,957	102,831	260,530
R-squared	0.9142	0.9148	0.9149	0.9173	0.9176	0.9299
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES

Notes: 18-month window, completed items. Standard errors clustered at the item level in parentheses. Columns (1)–(3) sequentially add competition mediators to the baseline price regression. Columns (4)–(6) restrict the sample to items with exactly 2, exactly 3, or 5+ participating firms, holding the extensive margin approximately constant. The persistence of the price effect across all subsamples rules out a pure headcount explanation for the cost reduction under open tenders; it does not, on its own, identify the intensive margin structurally. Firm count is itself an equilibrium outcome, so “same N in both regimes” does not guarantee the same bidder identities or cost distribution. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

11 Causal Forest

The [Athey, Tibshirani and Wager \(2019\)](#); [Wager and Athey \(2018\)](#) causal forest locates where treatment effects concentrate across the covariate space. I train the forest on the 18-month completed-items sample using log reference value, log quantity, class dummies, the sealed-bid indicator, and PBU type as inputs. Figure [OA.7](#) shows the variable-importance diagnostic: item quantity dominates with an importance of 0.486, roughly an order of magnitude above the next covariate. Figure [OA.8](#) then plots the group-average treatment effect (GATE) across quartiles of the estimated conditional-average treatment effect (CATE). The monotone decline from Q1 to Q4 is the heterogeneity signal that motivates the value-threshold counterfactual in Section 4.6.

Table OA.9: Causal Forest: Group Average Treatment Effects by CATE Quartile

	Q1 (lowest)	Q2	Q3	Q4 (highest)
GATE	0.3906 (0.9407)	-0.1143 (1.3634)	1.0482 (1.3738)	-0.8240 (0.8438)
ATE (full sample)	0.1251 (0.5779)			

Notes: Causal forest estimated with 2,000 honest trees on FWL-residualized outcomes. CATE quartiles defined by predicted individual treatment effects. Top variable importances: lquantidade: 0.486, convite: 0.183, fornec_estado_SP: 0.156, adm_dir: 0.079. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

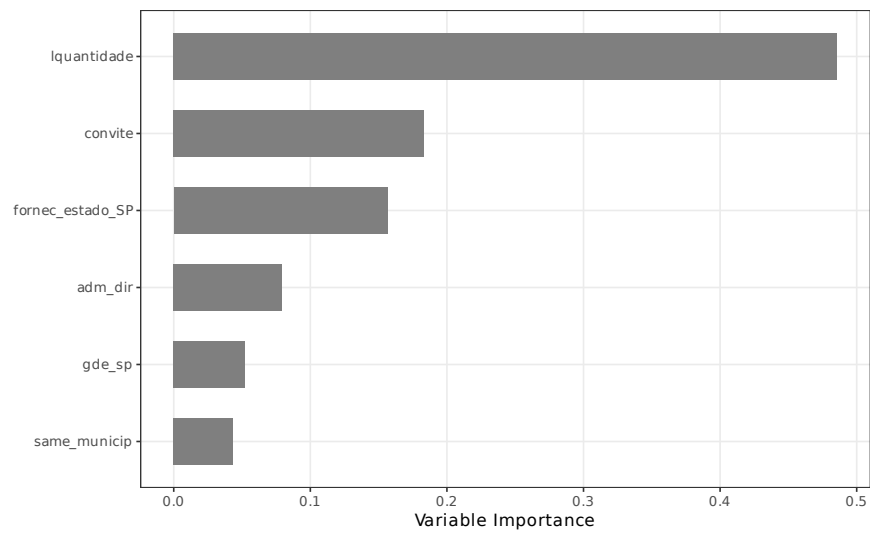


Figure OA.7: Causal Forest: Variable Importance for Treatment-Effect Heterogeneity

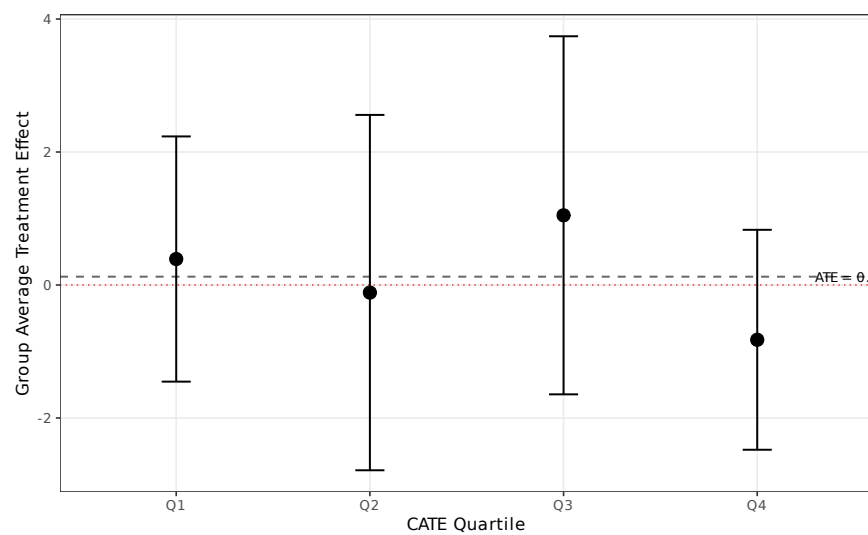


Figure OA.8: Causal Forest: Group Average Treatment Effects by CATE Quartile

12 Quantile Difference-in-Differences

The [Canay \(2011\)](#) quantile DiD on the residualized log-price distribution delivers a parallel check. Figure [OA.9](#) plots the coefficient at each decile. The effect is strongly negative at the lower quantiles ($\tau = 0.10$: -0.623 ; $\tau = 0.25$: -0.543) and turns positive at the upper tail ($\tau = 0.90$: $+0.445$). At the lower quantiles—where items are standardized and multiple capable suppliers exist—open competition drives prices down sharply; at the upper tail, items tend to be specialized with thinner markets, and the pattern reverses. The curve is the quantile-regression twin of the causal-forest GATE in Figure [OA.8](#), and of the residualized-QR diagnostic in Figure [33.3](#). Three different estimators deliver the same heterogeneity story.

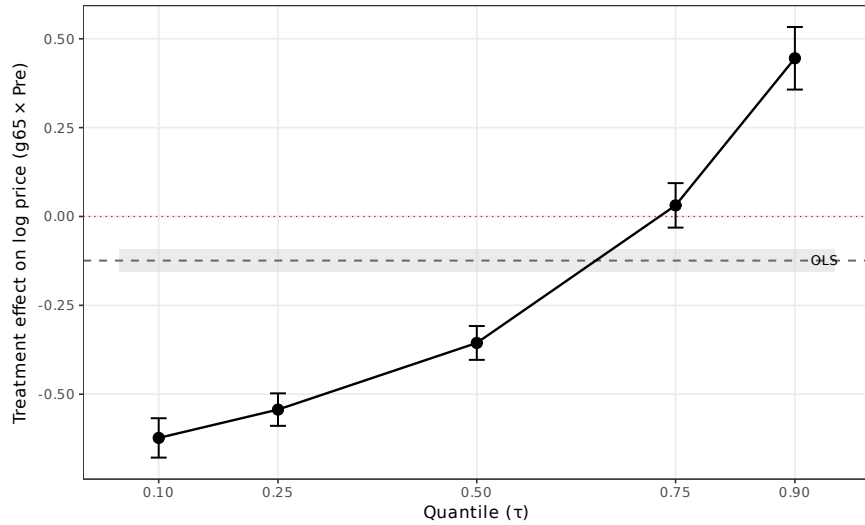


Figure OA.9: Quantile Difference-in-Differences: Treatment Effects Across the Price Distribution

13 Dose-Response by Item Quantity

Splitting the sample at the median of log item quantity ($= 3.18$) delivers the dose-response pattern referenced in the main text. Table [OA.11](#) reports the DiD coefficients separately on the high-quantity and low-quantity subsamples. The entry effect (log firms) scales roughly five-to-one between the two halves ($+0.125$ vs. $+0.025$), mirroring the causal-forest finding that item quantity is the dominant source of treatment-effect heterogeneity (variable importance $= 0.486$; Figure [OA.7](#)). The price effect is correspondingly larger in the low-quantity subsample because low-quantity items have a larger pool of standardized SME sellers but

Table OA.10: Quantile Difference-in-Differences: Treatment Effects Across the Price Distribution

	$\tau = 0.10$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.90$
$g65 \times Pre$	-0.6231*** (0.0283)	-0.5434*** (0.0233)	-0.3558*** (0.0243)	0.0313 (0.0319)	0.4451*** (0.0449)
OLS benchmark	-0.1241 (0.0165)				
Observations	100,000				
Group FE	YES				

Notes: Quantile regression estimated at each τ using the Canay (2011) two-step estimator with group FE (78 levels). 18-month window, completed items. OLS benchmark estimated on identical sample with group FE. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

a smaller non-SME pool that the SME-only rule removes, so the competition loss per removed non-SME bidder is bigger; high-quantity items absorb more of the extensive-margin hit through firm count itself.

Table OA.11: Dose-Response: Effects by Item Quantity

	Log prices		Log firms	
	Base	PBU FE	Base	PBU FE
<i>Panel A: High quantity (above median)</i>				
$g65 \times Pre$	-0.0963*** (0.0153)	-0.1037*** (0.0149)	0.1248*** (0.0094)	0.1393*** (0.0093)
Observations	348,363	348,363	402,194	402,194
<i>Panel B: Low quantity (below median)</i>				
$g65 \times Pre$	-0.1220*** (0.0129)	-0.1189*** (0.0122)	0.0251*** (0.0086)	0.0268*** (0.0084)
Observations	301,351	301,351	371,069	371,069

Notes: 18-month window. The sample is split at the median of log quantity (3.18). High-quantity items are more likely to attract large non-SME suppliers, so the SME restriction is more binding for these items. This split aligns with the causal forest finding that item quantity is the dominant source of treatment effect heterogeneity (variable importance = 0.486). SE clustered at item level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

14 Within-Group-65: Medications vs. Non-Pharma

Table OA.12 provides a split-sample complement to the interaction specification in Table OA.15. Restricting Group 65 to class 6531 (medications) yields a price coefficient of -0.166 (-0.174 with PBU FE); restricting Group 65 to the non-pharma classes yields -0.099 (-0.097 with PBU FE). The split-sample ratio is again close to 2:1, consistent with the triple-interaction specification in Table OA.15.

Table OA.12: Within-Group-65 Heterogeneity: Medications vs. Other Medical Supplies

	Log prices		Log firms	
	Base	PBU FE	Base	PBU FE
<i>Panel A: Medications (class 6531)</i>				
$g65_{med} \times Pre$	-0.1658^{***}	-0.1744^{***}	0.1516^{***}	0.1605^{***}
	(0.0174)	(0.0172)	(0.0100)	(0.0101)
Observations	573,080	573,080	678,547	678,547
<i>Panel B: Other medical supplies (non-6531)</i>				
$g65_{other} \times Pre$	-0.0992^{***}	-0.0973^{***}	0.0444^{***}	0.0504^{***}
	(0.0092)	(0.0086)	(0.0057)	(0.0057)
Observations	593,920	593,920	705,616	705,616

Notes: 18-month window. Panel A restricts group 65 to class 6531 (medications, subject to CMED pharmaceutical price regulation), using all non-group-65 items as the comparison. Panel B restricts group 65 to non-medication medical supplies (hospital equipment, dental supplies, etc.). If the price effect were driven by differential pharmaceutical inflation rather than competition, it should appear primarily in Panel A. SE clustered at item level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

15 Alternative Gelbach Parametrization

Table OA.13 reports an alternative Gelbach-style parametrization of the mediation exercise in the main text: columns (4)–(6) add log firms, the SME-winner indicator, and both mediators sequentially to the baseline price regression under item fixed effects only (no month FE). The direct price coefficient on $g65 \times Pre$ grows in magnitude under each mediator, consistent with the negative-weighting suppressor structure documented in Table OA.6 above: the two observable channels offset one another in net, leaving the direct coefficient larger than the short-regression baseline.

Table OA.13: Mechanism Decomposition: Entry Margin and Winner Composition

	(1) Log price	(2) Log firms	(3) SME winner	(4) Price + firms	(5) Price + SME	(6) Price + both
$g65 \times Pre$	-0.1087*** (0.0120)	0.0978*** (0.0070)	-0.1001*** (0.0064)	-0.1395*** (0.0099)	-0.1159*** (0.0098)	-0.1227*** (0.0101)
Log firms				0.1087*** (0.0106)		0.1099*** (0.0106)
SME winner					0.0390*** (0.0037)	0.0438*** (0.0037)
Observations	649,714	773,263	649,714	648,616	649,714	648,616

Notes: 18-month window, item FE, SE clustered at item level. Columns (1)–(3): DiDiR for each outcome separately. Column (4): price regression adding log firms as control (mediation by competition). Column (5): price regression adding SME winner indicator (mediation by composition). Column (6): price regression with both mediators. The attenuation of the $g65 \times Pre$ coefficient from (1) to (4)–(6) measures the fraction of the price effect explained by each channel. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Part III

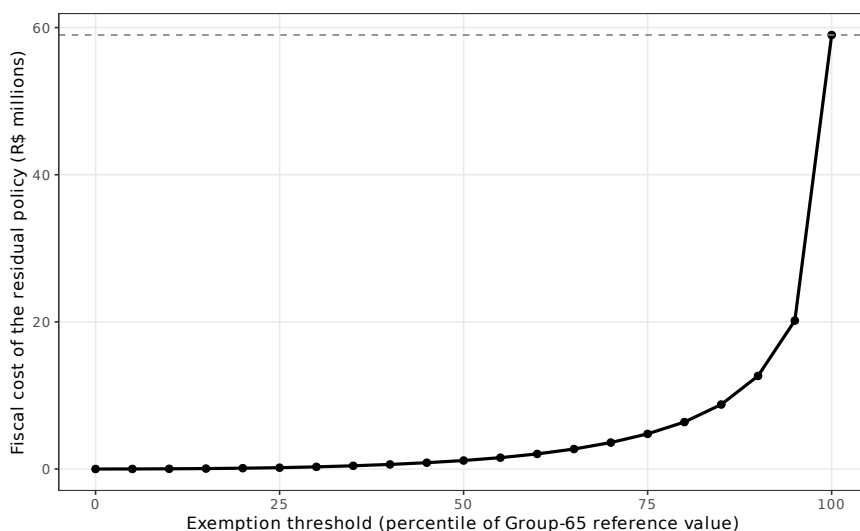
Policy Design and Extensions

16 Value-Threshold Counterfactual

Table OA.14 reports the per-quartile DiD coefficients used in the value-threshold counterfactual simulation. The simulation splits Group-65 pre-period items into quartiles of reference value (cutoffs R\$500 / US\$143, R\$1,875 / US\$536, R\$7,623 / US\$2,180 (at R\$3.50 per US dollar)), estimates the DiD within each quartile, and computes the fiscal cost recovered by exempting the top k quartiles. Because 92% of pre-period Group-65 procurement value concentrates in the top quartile, exempting Q4 alone recovers 90% of the counterfactual baseline fiscal cost.

Table OA.14: Counterfactual Policy Designs (reprinted)

Figure 16 reprints the residual-fiscal-cost schedule as a function of the exemption percentile, complementing the Pareto-frontier view of Figure ?? in the main text. Both figures deliver the same message through different axes: the first reads the marginal cost of preserving the SME-only rule at each percentile, while the second reads the cumulative cost recovered by exempting everything above a percentile.



Residual Fiscal Cost as a
Function of the Exemption Percentile

17 CMED-Regulated Pharmaceuticals vs. Non-Pharma Medical Supplies

The pharma decomposition splits Group 65 into class 6531 (medications subject to CMED price ceilings) and the remaining classes (non-pharma medical supplies). Split-sample DiD estimates—with controls included in both subsamples so that $\hat{\beta}$ is identified against the never-treated groups in each case—are -0.156 for pharma and -0.075 for non-pharma; the differential is confirmed by an interaction specification within the full 18-month sample. Pharma bears 60% of the total fiscal cost despite 42% of the procurement value, because the per-unit distortion is roughly double that of the non-pharma subsegment. A direct look at prices (Figure ?? in the main text) shows that 96% of pharma transactions clear below the CMED-referenced cap, so the cap is slack and the SME-only rule compresses competition within the residual below-cap range. Implementation details are in `scripts/24_pharma.R`.

18 Real Prices (IPCA-Deflated)

19 Extensive Margin and Procurement Efficiency

20 Heterogeneity by PBU Type

21 SME Winner Composition

Figure [OA.10](#) displays the monthly share of completed items where the winning firm is self-classified as an SME.

Table OA.15: Pharmaceuticals (CMED-Regulated) vs. Non-Pharma Medical Supplies

	Pharma (class 6531) split sample	Non-pharma (other 65) split sample	Interaction full sample
<i>Log prices</i>			
$g65 \times Pre$	-0.1556*** (0.0192)	-0.0746*** (0.0101)	-0.0785*** (0.0103)
$g65 \times Pre \times Pharma$	— —	— —	-0.0757*** (0.0191)
<i>Log firms</i>			
$g65 \times Pre$	0.1649*** (0.0109)	0.0574*** (0.0067)	0.0589*** (0.0067)
$g65 \times Pre \times Pharma$	— —	— —	0.1063*** (0.0115)
<i>Log bids</i>			
$g65 \times Pre$	0.1502*** (0.0151)	0.0296*** (0.0105)	0.0301*** (0.0105)
$g65 \times Pre \times Pharma$	— —	— —	0.1192*** (0.0153)
<i>Distance (km)</i>			
$g65 \times Pre$	10.2758** (4.3598)	3.3392* (1.9651)	2.8388 (1.9656)
$g65 \times Pre \times Pharma$	— —	— —	7.2164 (4.6283)
Obs. (Log prices)	573,080	593,920	649,714
Obs. (Log firms)	678,547	705,616	773,263
Obs. (Log bids)	678,547	705,616	773,263
Obs. (Distance)	573,080	593,920	649,714
Item FE + PBU FE	YES	YES	YES
<i>Implied fiscal cost (pre-period, nominal)</i>			
Value of Group-65 procurement (R\$M)	293.1	395.9	689.0
Fiscal cost (R\$M)	42.2	28.4	70.7
Share of total cost	60%	40%	100%

Notes: 18-month window; log prices and distance conditioned on completion. Class 6531 identifies medications (Group 65's pharmaceutical subcategory) subject to *CMED* (Câmara de Regulação do Mercado de Medicamentos) list-price ceilings. Remaining Group-65 classes (non-pharma medical supplies) are unregulated. The first two columns estimate the headline DiD separately within each subsample. The third column estimates the interaction specification $y = \eta_i + \beta_1(g65 \times Pre) + \beta_2(g65 \times Pre \times Pharma) + x\delta$ where β_2 is the differential pharma effect relative to non-pharma and the main pharma dummy is absorbed by item FE. SEs clustered at the item level. The implied fiscal cost applies the class-specific $\hat{\beta}$ on log prices to the pre-period Group-65 procurement value within that class. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table OA.16: Real Prices (IPCA-deflated, log): Pre-switch-period effect on group 65

	(1)	(2)	(3)	(4)	(5)	(6)
	6-month	6-month	12-month	12-month	18-month	18-month
$g65 \times Pre$	-0.1417*** (0.0146)	-0.1482*** (0.0144)	-0.1079*** (0.0133)	-0.1077*** (0.0129)	-0.1087*** (0.0120)	-0.1133*** (0.0117)
Sealed bids	-0.1409*** (0.0086)	-0.2176*** (0.0130)	-0.1380*** (0.0077)	-0.1991*** (0.0122)	-0.1455*** (0.0077)	-0.2018*** (0.0114)
lquantity	-0.2616*** (0.0082)	-0.2986*** (0.0089)	-0.2634*** (0.0075)	-0.2974*** (0.0080)	-0.2617*** (0.0073)	-0.2949*** (0.0077)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.1919	0.2132	0.1943	0.2120	0.1947	0.2118
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Constant absorbed by fixed effects.

Table OA.17: Extensive Margin: Tender Completion Rate

	(1)	(2)	(3)	(4)	(5)	(6)
	6-month	6-month	12-month	12-month	18-month	18-month
$g65 \times Pre$	0.1165*** (0.0044)	0.1179*** (0.0044)	0.1244*** (0.0035)	0.1265*** (0.0036)	0.1072*** (0.0032)	0.1133*** (0.0032)
Sealed bids	0.0735*** (0.0025)	0.0607*** (0.0033)	0.0638*** (0.0020)	0.0486*** (0.0025)	0.0651*** (0.0018)	0.0544*** (0.0023)
lquantity	0.0224*** (0.0007)	0.0256*** (0.0007)	0.0258*** (0.0006)	0.0284*** (0.0006)	0.0251*** (0.0005)	0.0274*** (0.0005)
Observations	304,392	304,392	609,299	609,299	901,506	901,506
R-squared	0.0181	0.0177	0.0211	0.0209	0.0195	0.0194
Group FE	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: DV = 1 if item was successfully purchased, 0 otherwise. All items included (not filtered to completed). Group FE used instead of item FE. Standard errors clustered at the item level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table OA.18: Price Efficiency: Final Price / Reference Price (%)

	(1)	(2)	(3)	(4)	(5)	(6)
	6-month	6-month	12-month	12-month	18-month	18-month
$g65 \times Pre$	-0.0563*** (0.0118)	-0.0596*** (0.0113)	-0.4259 (0.3813)	-0.4723 (0.4260)	-0.2921 (0.2623)	-0.3237 (0.2949)
Sealed bids	-0.0622*** (0.0027)	-0.0498*** (0.0044)	0.0279 (0.0850)	0.0926 (0.1385)	-0.0057 (0.0531)	0.0330 (0.0835)
lquantity	-0.0151*** (0.0018)	-0.0157*** (0.0018)	0.1226 (0.1381)	0.1351 (0.1496)	0.0717 (0.0876)	0.0777 (0.0939)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.0037	0.0023	0.0001	0.0001	0.0000	0.0000
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Constant absorbed by fixed effects.

Table OA.19: Heterogeneous Effects by PBU Type (18-month window)

	Log prices		Log firms	
	Base	+PBU FE	Base	+PBU FE
$g65 \times Pre$	-0.0718*** (0.0231)	-0.0697*** (0.0230)	-0.0146 (0.0113)	0.0567*** (0.0120)
Direct admin.	0.0425*** (0.0084)	—	0.0421*** (0.0044)	—
$g65 \times Pre \times$ Direct admin.	-0.0655*** (0.0236)	-0.0685*** (0.0227)	0.1175*** (0.0117)	0.0475*** (0.0122)
Observations	649,714	649,714	773,263	773,263
R-squared	0.1936	0.2108	0.2065	0.1632
Item FE	YES	YES	YES	YES
PBU FE	NO	YES	NO	YES

Notes: 18-month window. Direct admin. is 1 for PBUs under direct administration, 0 otherwise. Standard errors clustered at the item level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table OA.20: Winner Composition: SME Winner (binary)

	(1)	(2)	(3)	(4)	(5)	(6)
	6-month	6-month	12-month	12-month	18-month	18-month
$g65 \times Pre$	-0.0743*** (0.0072)	-0.0857*** (0.0072)	-0.1274*** (0.0067)	-0.1321*** (0.0066)	-0.1001*** (0.0064)	-0.1162*** (0.0063)
Sealed bids	0.4038*** (0.0042)	0.3767*** (0.0047)	0.4642*** (0.0036)	0.4324*** (0.0036)	0.4211*** (0.0031)	0.3856*** (0.0029)
lquantity	0.0081*** (0.0010)	0.0108*** (0.0008)	0.0042*** (0.0008)	0.0076*** (0.0006)	0.0016** (0.0007)	0.0058*** (0.0005)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.1237	0.0986	0.1619	0.1216	0.1417	0.0981
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Constant absorbed by fixed effects.

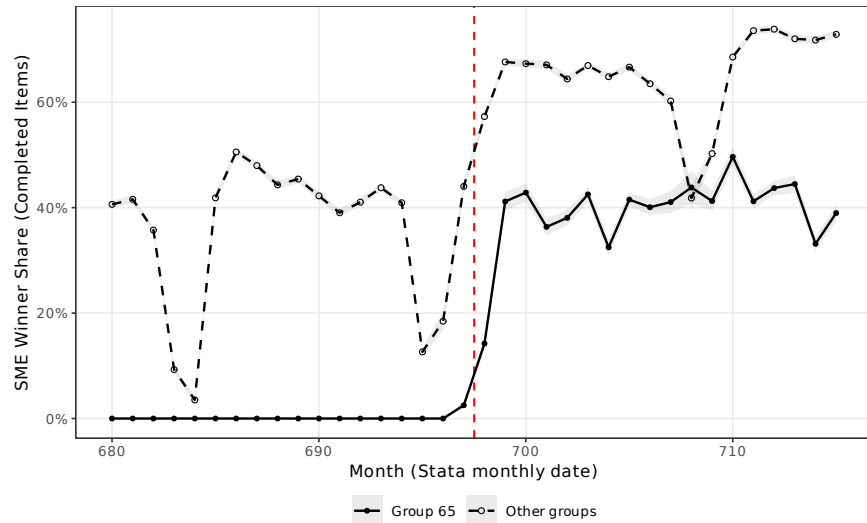


Figure OA.10: SME Winner Share (Completed Items) by Month

Part IV

RAIS Matched Employer-Employee Linkage

Brazil’s *Relação Anual de Informações Sociais* (RAIS) is a matched employer-employee registry covering the universe of formal-sector employment relationships. This part links the winning firms in BEC to their RAIS profiles at the CNPJ-raiz (8-digit firm identifier) level, using the establishment-level (ESTB) files for 2011–2017. Of 17,857 distinct winning CNPJ raízes in the Paper-2 sample, 14,755 match RAIS 2017 nationwide (82.6% coverage); within Sao Paulo state alone, 12,533 match (70.2%). The link is produced by `scripts/05_link_rais.py`, a DuckDB-based ETL.

22 Validation of the Self-Declared ME/EPP Label

The BEC data includes a self-declared `fornec_enquad` (firm-enquadramento) variable. Table [OA.21](#) tests whether the headline DiD is sensitive to the label’s classification noise by restricting the estimation sample to firms with (i) a RAIS match, (ii) active employment at most 49 (ME/EPP ceiling), (iii) active employment at most 9 (micro ceiling), and (iv) active employment strictly below 100 (excluding the top of the RAIS size distribution). The price coefficient is essentially invariant across these subsamples.

Table OA.21: RAIS-Validated Winner Subsamples

23 Winner Composition: RAIS-Validated SME

Table [OA.22](#) uses a RAIS-validated SME flag (winner active employment ≤ 49) as the dependent variable. The DiD on this outcome estimates how the SME-only rule shifts the probability of a genuinely small winner; the mechanism prediction is a positive post-policy shift.

Table OA.22: Winner Composition: RAIS-Validated SME

Table OA.23: Winner Composition: Alternative RAIS Size Thresholds (18-month window)

	Micro (≤ 9 vínculos)		Not large (< 100 vínculos)	
	Base	+PBU FE	Base	+PBU FE
$g65 \times Pre$	-0.3094*** (0.0072)	-0.3180*** (0.0072)	-0.1986*** (0.0067)	-0.2019*** (0.0068)
Observations	649,714	649,714	649,714	649,714
R-squared	0.0246	0.0247	0.0156	0.0155
Item FE	YES	YES	YES	YES
PBU FE	NO	YES	NO	YES

Notes: Item-level binary outcome equal to one if the winner's CNPJ raiz matches RAIS 2017 (Brasil) with at most 9 (Micro) or fewer than 100 (Not large) active employment links summed across establishments. 18-month window, completed items. Compare to Table ?? (SME ≤ 49) and Table OA.20 (self-declared). Standard errors clustered at the item level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

24 Main DiD with RAIS Firm-Level Controls

Table OA.24 augments the headline DiD with RAIS firm-level controls— $\log(1 + \text{vínculos})$, $\log(1 + \text{firm age at 2018})$, and CNAE 2-digit fixed effects. The $g65 \times Pre$ coefficient is invariant, confirming that the price effect is not explained by observable differences in winner size, age, or sector.

Table OA.24: Main DiD with RAIS Firm-Level Controls

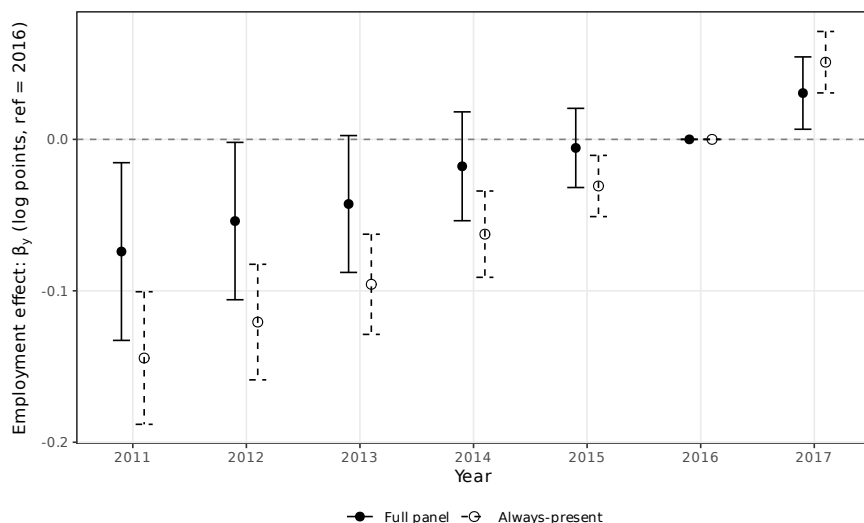
25 Enforcement Heterogeneity

Table OA.25 tests whether the price effect is larger in PBUs that were historically more exposed to non-SME winners. The exposure measure is the pre-period share of each PBU's winners with RAIS employment ≥ 100 . Weighted median is 6.9%; PBUs above median show a -0.049 ($p < 0.05$) larger price distortion.

Table OA.25: Enforcement Heterogeneity by PBU Exposure

26 Firm-Level Employment Pre-Trends (RAIS 2011–2017)

Figure 26 plots the event-study coefficients on $\log(1 + \text{vínculos})$ for treated firms (ever won a group-65 item in the BEC pre-period) versus controls (won only non-group-65 items in the same period). Pre-reference joint F -test is 1.46 ($p = 0.198$) in the balanced panel with zero-fill, failing to reject parallel pre-trends. The always-present subset (firms with ≥ 1 employment link every year 2011–2017) rejects more strongly; this reflects the selection of procurement winners on firm productivity and growth rather than a violation of the paper’s item-level identification.



Pre-Trends (RAIS 2011–2017)

Table OA.26: Pre-Trends Event Study

27 Firm Age Heterogeneity

Table OA.27 interacts $g65 \times Pre$ with a young-firm indicator (age at 2018 ≤ 9 years, the median of pre-period winners). The price effect is uniform across age within the ME/EPP-eligible pool; the apparent differences reported by naive age splits are driven by firm-size composition (see Section OA.27).

Table OA.27: Heterogeneity by Winner Firm Age

28 CNAE 2-Digit Sector Heterogeneity

Table OA.28 re-estimates the DiD within each of the top-6 CNAE 2.0 divisions (wholesale 46, retail 47, vehicle wholesale 45, chemicals 20, food manufacturing 10, other manufacturing 32) and reports the coefficient on $g65 \times Pre$.

Table OA.28: Heterogeneity by Winner CNAE 2-Digit Sector

29 Winner Age $\times$ Size Quadrants

Table OA.29 crosses winner age (young/mature, median split at 9 years) with RAIS-observed size (SME/large, cut at the 49-vínculos ME/EPP ceiling) to form four quadrants. Within eligible firms (SME), the price effect is essentially identical for young and mature firms (-0.129 vs -0.128)—the rent transfer from the SME-only rule is uniform within eligibility and concentrated on the geographic/competition margins rather than on firm maturity *per se*.

Table OA.29: Heterogeneity by Age $\times$ Size Quadrants

Part V

Complementary Estimators

30 Synthetic Control

As an external check that does not lean on the DiD framing, I build a synthetic control for Group 65 using the 76 never-treated product groups as donors (Abadie, Diamond and Hainmueller, 2010; Ben-Michael, Feller and Rothstein, 2021). Donor weights are chosen to match the Group-65 pre-period trajectory in log prices. Figure OA.11 overlays actual Group-65 log prices against the synthetic counterfactual: the pre-period fit is near-exact, and a visible gap opens after March 2018. Figure OA.12 plots that gap directly. Figure OA.13 reports the in-space placebo distribution of post-treatment ATTs across donor groups—the Group-65 post-switch gap sits in the extreme tail of the placebo distribution, so the effect is not a product of random noise.

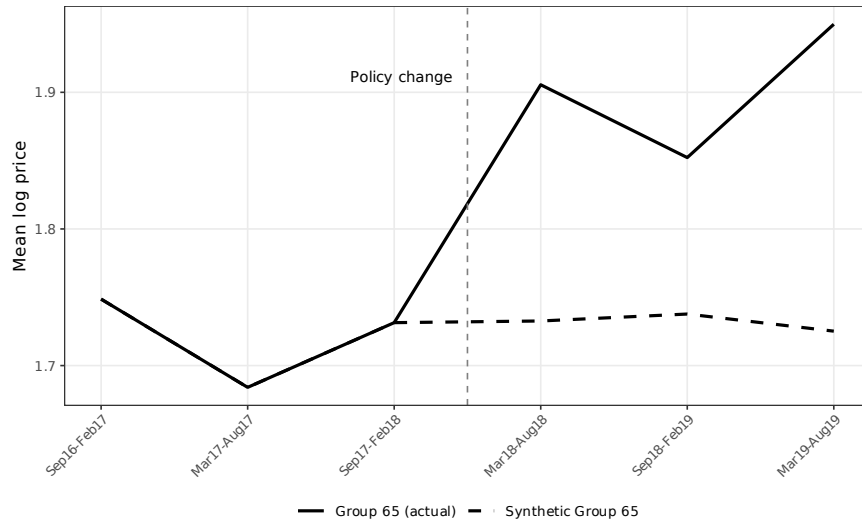


Figure OA.11: Synthetic Control: Group 65 (Actual) vs. Synthetic Group 65

31 HonestDiD Sensitivity Analysis

Figure 31 reprints the Rambachan and Roth (2023) sensitivity diagnostic from the main appendix, which reports robust 95% confidence intervals for each outcome under post-treatment deviations from parallel trends up to $\bar{M}$ times the largest pre-period violation. Under the two-way fixed-effects specification, the log-price robust CI narrowly excludes zero near $\bar{M} = 0$

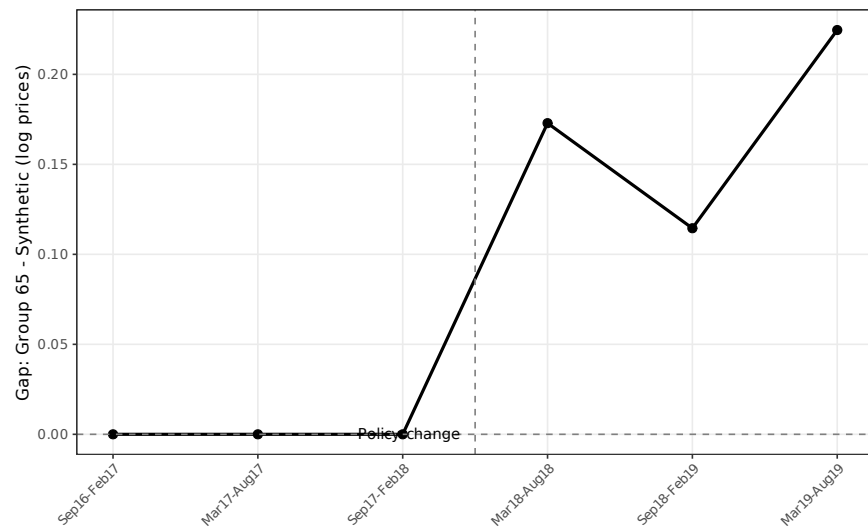


Figure OA.12: Synthetic Control: Gap Between Group 65 and Synthetic (Log Prices)

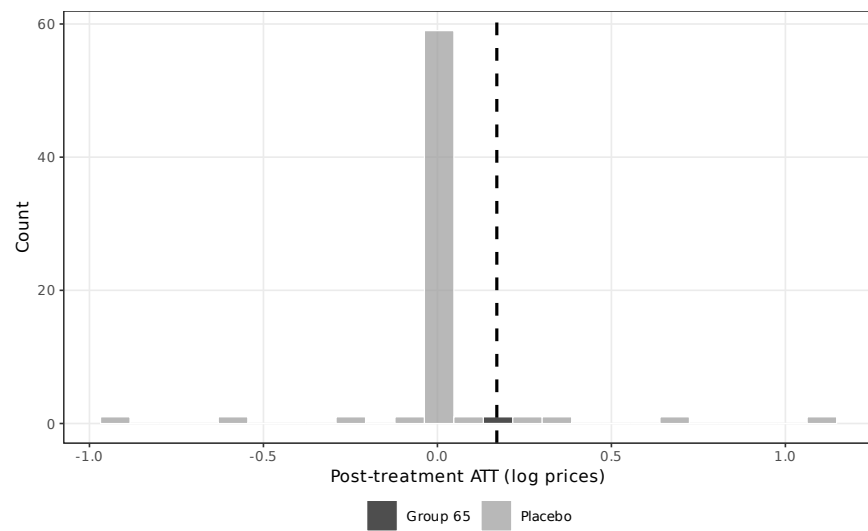


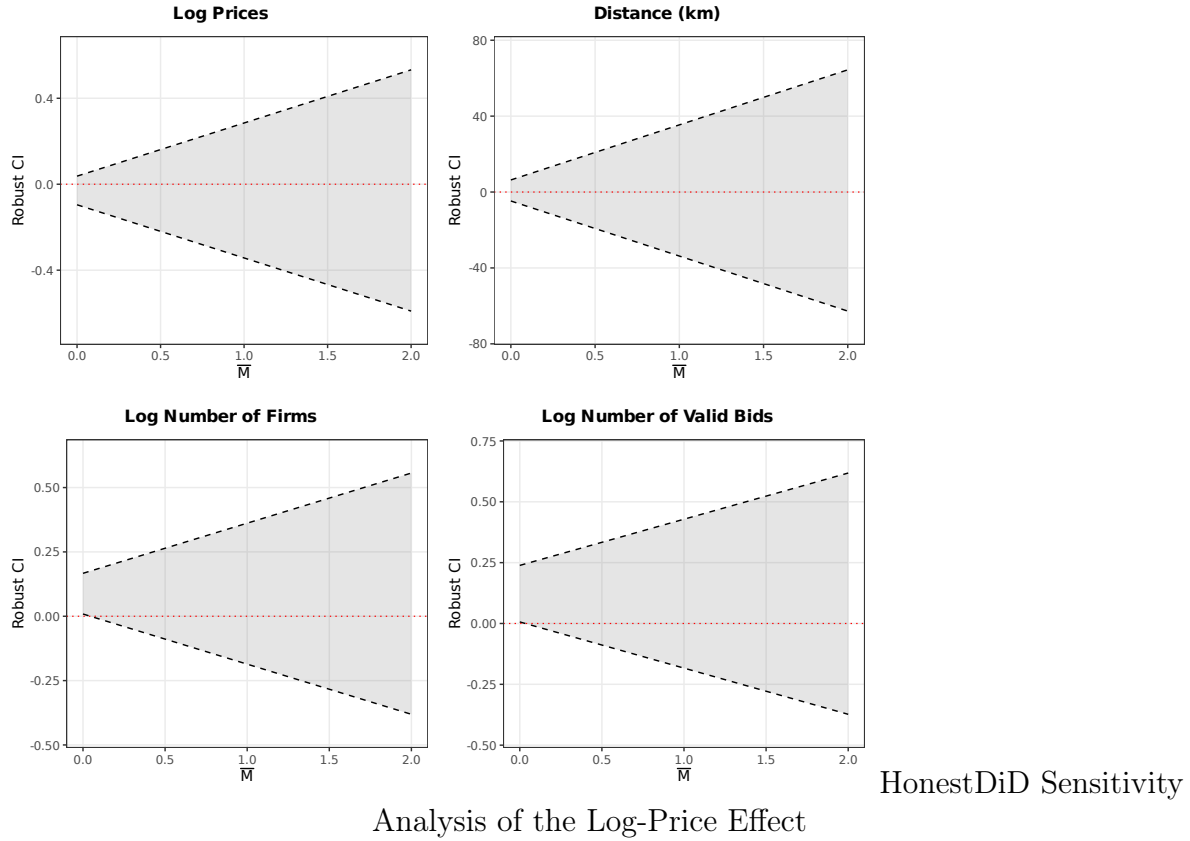
Figure OA.13: Synthetic Control: In-Space Placebo Distribution of Post-Treatment ATTs

Table OA.30: Synthetic Control: Summary Results

<i>Panel A: Gaps</i>	
Mean pre-treatment gap (semesters 1–3)	0.0000
Mean post-treatment gap (semesters 4–6)	0.1707
Post-treatment ATT (mean)	0.1707
Placebo rank p -value	0.103 (7/68)
L2 imbalance	0.0000
<i>Panel B: Top donor weights ($> 1\%$)</i>	
Product group	Weight
Group 40	0.044
Group 75	0.099
Group 79	0.382
Group 89	0.436
Other groups	0.039
Balanced groups	69
Semesters	6

Notes: Augmented synthetic control with Ridge augmentation. Outcome: mean log price at the group $\times$ semester level. Treatment date: March 2018 (group 65 loses open-tender exemption). Pre-treatment gap measures how much lower group 65 prices were under open tenders relative to the synthetic counterfactual. Post-treatment gap near zero validates the parallel trends assumption. Inference via conformal method.

but includes zero beyond approximately $\bar{M} \approx 0.1$. Read conservatively, this means the price effect is robust only to small pre-trend violations; read together with the clean price placebo, the event study, and the multi-estimator convergence, it is an honest caveat rather than a rejection. The worst-case HonestDiD bound is a stricter test than the point estimate supports in isolation.



32 Lee (2009) Bounds

Table OA.31: Lee (2009) Bounds: Sample Selection Correction

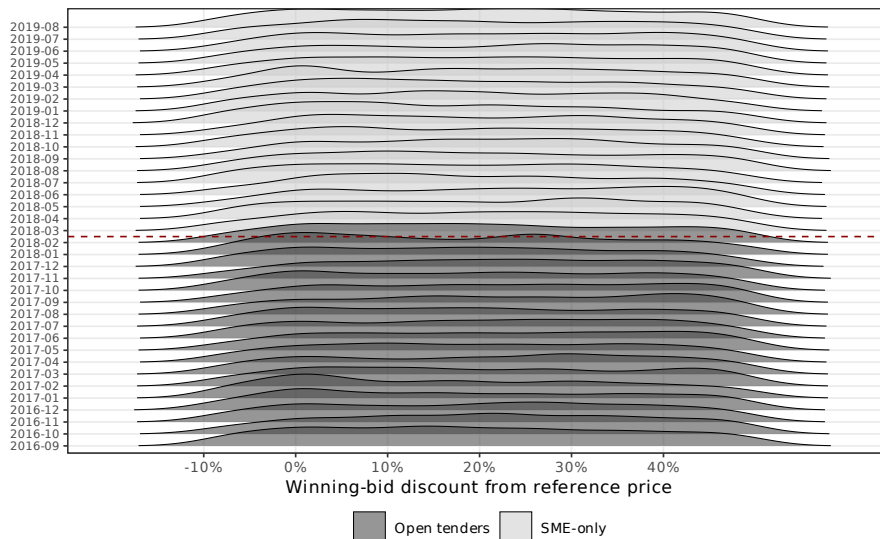
	Log prices		Distance (km)	
	Lower bound	Upper bound	Lower bound	Upper bound
$g65 \times Pre$	-0.0460*** (0.0131)	-0.1338*** (0.0114)	27.0609*** (2.3804)	-19.0176*** (2.3672)
Observations	625,816	625,814	625,814	625,814
R-squared	0.1886	0.1951	0.0015	0.0017
Trimming proportion		0.0882		
Item FE	YES	YES	YES	YES

Notes: Lee (2009) bounds correct for sample selection arising from treatment effects on completion rates. 18-month window. Lower/upper bounds obtained by trimming the outcome distribution in the excess-selected cell. DiD in completion rate: -0.0882. Standard errors clustered at the item level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

33 Complementary Visualizations

33.1 Monthly Bid-Discount Densities

Figure 33.1 stacks the monthly kernel densities of the winning-bid discount for Group-65 items through the 18-month window. The regime change lands visibly in March 2018: post-cutoff months bunch near a zero discount, while pre-cutoff months keep meaningful mass on discounts of twenty percent or more.



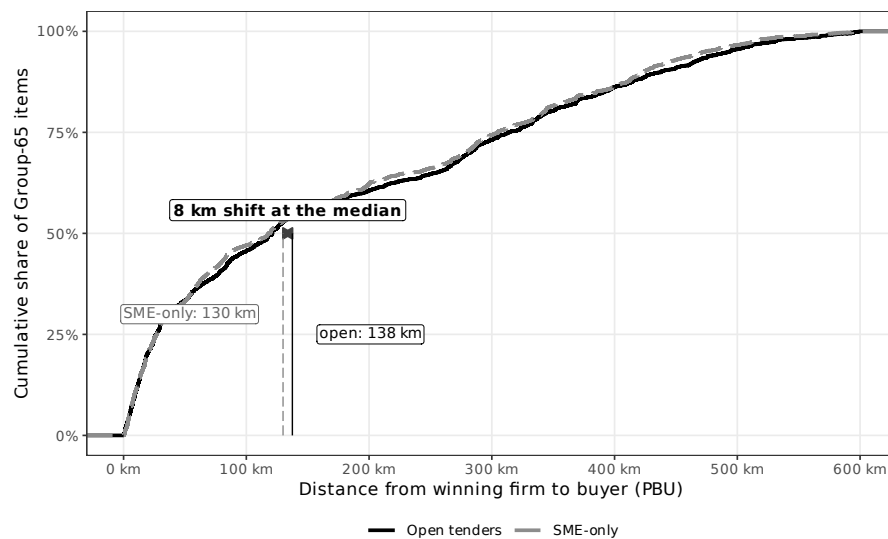
Monthly density of
Group-65 winning-bid discount, 2016–2019

33.2 Winner-to-PBU Distance Density

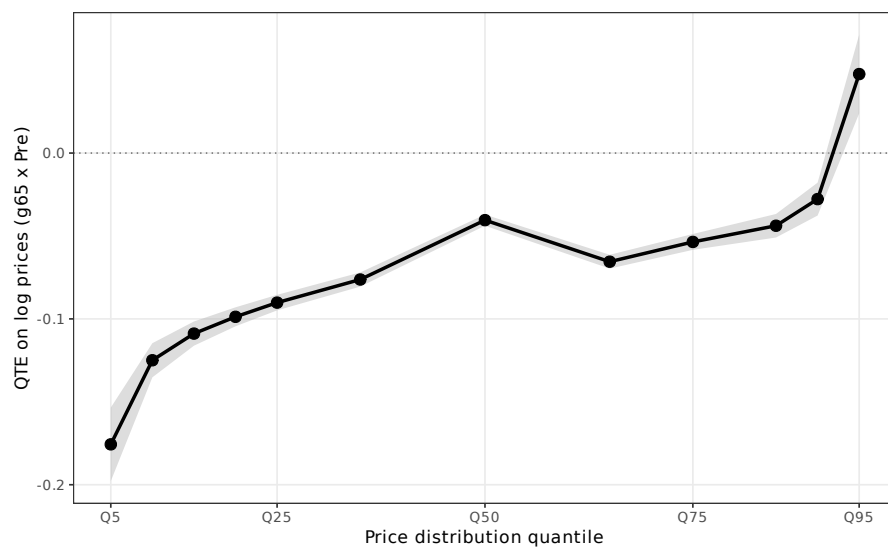
Figure 33.2 plots the empirical CDF of the kilometer distance from the winning firm to the buying PBU, by regime. The 50% line sits 8 km further out under open tenders—the geographic counterpart of the conditional-mean distance estimates in Table ??.

33.3 Quantile Treatment Effect on Log Prices

Figure 33.3 reports the quantile treatment effect of $g65 \times Pre$ on log prices across twelve quantiles of the within-item price distribution, computed via residualized quantile regression. The effect is strongly negative at the lower quantiles and fades toward zero at the upper tail—the same heterogeneity pattern documented in Section 4.4 and in the Canay quantile DiD of Section OA.B.



Winner-to-PBU distance
density by regime, completed Group-65 items



Quantile treatment effect
on log prices, residualized quantile regression

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